

Computer Vision

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2. Calibrated vs. Uncalibrated Reconstruction

Let T be any non-singular 4×4 matrix, representing a projective transformation of 3D space.

Define new 3D points and new cameras by

$$X_j' = TX_j, P_i' = P_i T^{-1}$$

Then each image point satisfies

$$x_{ij} \sim P_i X_j = P_i T^{-1} T X_j = P_i' X_j'$$

so the reprojected image coordinates x_{ij} are identical for the pair (P_i, X_j) and for the transformed pair (P_i', X_j')

If X denotes the set (or matrix) of all reconstructed 3D points, then for any projective transformation T of 3D space the pair $(\{P_i T^{-1}\}_i, \{TX_j\}_j)$ is a different 3D configuration and camera set that yields exactly the same image measurements. Therefore, when cameras are uncalibrated (no constraints on the internal parameters), the structure and motion can only be recovered up to an arbitrary 3D projective transformation T ; additional information (calibration or metric constraints) is required to upgrade this projective reconstruction to a metric or Euclidean one.

with calibrated cameras the reconstruction is metric/Euclidean up to a single global similarity transform, meaning that distances are known up to scale and angles are preserved, but there remains the unavoidable ambiguity of choosing the world coordinate frame and overall scale.

3. Camera Calibration

Matrix A is a uniform scaling in the image plane by the factor $1/f$ in both x and y , which converts pixel coordinates into normalized image coordinates where the effective focal length is 1. Matrix B is a translation in the image plane by $(-x_0, -y_0)$, which shifts the origin from the upper-left image corner to the principal point (x_0, y_0) , i.e., to where the optical axis hits the image. Therefore $K^{-1} = AB$ first recenters the coordinates at the principal point (via B), then rescales them so that the focal length becomes 1 (via A), mapping pixel coordinates to normalized camera coordinates.

Applying K^{-1} to image points converts pixel coordinates into normalized camera coordinates: it shifts the origin to the principal point and scales by $1/f$, so that coordinates are measured in units of focal length and the principal point moves to $(0,0)$. A point at distance f from the principal point (in pixels) ends up at unit distance 1 from the origin in the corresponding direction in normalized coordinates.

For this camera,

$$K = \begin{pmatrix} 360 & 0 & 360 \\ 0 & 360 & 200 \\ 0 & 0 & 1 \end{pmatrix}, \quad K^{-1} = \begin{pmatrix} 1/360 & 0 & -1 \\ 0 & 1/360 & -5/9 \\ 0 & 0 & 1 \end{pmatrix}. [attached_file : 1]$$

Normalizing $(0,200)$ gives

$$\hat{x}_1 = K^{-1} \begin{pmatrix} 0 \\ 200 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

so the normalized point is $(-1,0)$.

Normalizing (720,200) gives

$$\hat{x}_2 = K^{-1} \begin{pmatrix} 720 \\ 200 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

so the normalized point is (1,0).

The viewing rays corresponding to normalized points $(-1,0)$ and $(1,0)$ (on the plane $Z = 1$) are rays through the camera center and the 3D points $(-1,0,1)$ and $(1,0,1)$, respectively.

Their directions are proportional to vectors $(-1,0,1)$ and $(1,0,1)$; the angle θ between them satisfies

$$\cos \theta = \frac{(-1, 0, 1) \cdot (1, 0, 1)}{\|(-1, 0, 1)\| \|(1, 0, 1)\|} = \frac{0}{\sqrt{2}\sqrt{2}} = 0,$$

hence $\theta = 90^\circ$

4. RQ Factorization and Computation of K

Use that for a calibrated camera matrix $P = K[Rt]$ with K upper triangular and positive diagonal, the left 3×3 block is KR .

From the formula in the problem,

$$KR = \begin{pmatrix} aR_1^T + bR_2^T + cR_3^T \\ dR_2^T + eR_3^T \\ fR_3^T \end{pmatrix},$$

so the third row of KR is simply fR_3^T

In the given matrix

$$P = \begin{pmatrix} 2400\sqrt{2} & 0 & 800\sqrt{2} & 4000 \\ 700\sqrt{2} & 2800 & -700\sqrt{2} & 4900 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} & 3 \end{pmatrix},$$

the left 3×3 block is

$$A = \begin{pmatrix} 2400\sqrt{2} & 0 & 800\sqrt{2} \\ 700\sqrt{2} & 2800 & -700\sqrt{2} \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix} . [attached_file : 1]$$

Thus the third row of A is

$$fR_3^T = \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right) . [attached_file : 1]$$

Since R_3 must be a unit vector (rows of an orthogonal matrix have unit norm),

$$f = \left\| \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right) \right\| = \sqrt{\left(\frac{\sqrt{2}}{2} \right)^2 + 0^2 + \left(-\frac{\sqrt{2}}{2} \right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1. \square \square$$

Therefore

$$R_3^T = \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right), \quad \text{so} \quad R_3 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ -\frac{\sqrt{2}}{2} \end{pmatrix}$$

and the focal length is $f = 1$.

From the previous part the third row of KR is

$$fR_3^T = \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right)$$

so with $f = 1$ one has

$$R_3^T = \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right). [attached_file : 6]$$

The second row of the left 3×3 block of P is

$$v = \left(700\sqrt{2}, 2800, -700\sqrt{2} \right)$$

and by the given formula this equals $dR_2^T + eR_3^T$

Because R_2 and R_3 are orthonormal, the decomposition coefficients of v in this basis are found by inner products:

$$e=v \cdot R_3=1400, v_{\perp}=v-eR_3=(0,2800,0).[attachedfile:6]$$

The orthogonal component has norm

$$d=\|v_{\perp}\|=2800$$

and its direction is $R_2^T = v_{\perp}/d = (0,1,0)$, so

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad d = 2800, \quad e = 1400.[attachedfile : 6]$$

From earlier steps the rotation rows are

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad R_3 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ -\frac{\sqrt{2}}{2} \end{pmatrix}.[attachedfile : 5][attachedfile : 6]$$

The first row of the left 3×3 block of P is

$$v=(24002,0,8002).[attachedfile:5]$$

Using the decomposition $v = aR_1^T + bR_2^T + cR_3^T$ with orthonormal R_1, R_2, R_3 :

$$b=v \cdot R_2=0, c=v \cdot R_3=1600.[attachedfile:7]$$

The remaining component is

$$v_{\text{rem}}=v-bR_2^T-cR_3^T=(16002,0,16002),[attachedfile:7]$$

so

$$a = \|v_{\text{rem}}\| = 3200, \quad R_1^T = \frac{v_{\text{rem}}}{a} = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2} \right).[attachedfile : 7]$$

Thus the intrinsic matrix is

$$K = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix} = \begin{pmatrix} 3200 & 0 & 1600 \\ 0 & 2800 & 1400 \\ 0 & 0 & 1 \end{pmatrix}, [attached_file : 6][attached_file : 7]$$

so the focal length in x is $f_x = 3200$, in y is $f_y = 2800$, the skew is $s = b = 0$, and the principal point is $(c, e) = (1600, 1400)$.

The aspect ratio is $f_y/f_x = 2800/3200 = 0.875$

5. Direct Linear Transformation DLT

Because the normalized image points satisfy $\tilde{\mathbf{x}} \sim N\mathbf{x}$ and also $\tilde{\mathbf{x}} \sim \tilde{P}\mathbf{X}$, while the original points satisfy $\mathbf{x} \sim P\mathbf{X}$, one has

$$\tilde{P}\mathbf{X} \sim N\mathbf{P}\mathbf{X} \quad \Rightarrow \quad \tilde{P} \sim N\mathbf{P}.$$

Thus the camera matrix for the original (unnormalized) image coordinates is

$$P \sim N^{-1}\tilde{P}$$

i.e. multiply the normalized camera matrix on the left by N^{-1}